WCAG 2.2 Level A & AA: A Product Owner's Guide to Accessible Digital Products

Part 1: Making Your Product Accessible: A Product Owner's Introduction to WCAG 2.2

Section 1.1: What is WCAG 2.2 and Why It's Critical for Your Product's Success

The Web Content Accessibility Guidelines, commonly known as WCAG, represent the global benchmark for creating accessible digital experiences. Developed by the World Wide Web Consortium (W3C), an international community that develops open standards to ensur e the long-term growth of the Web, WCAG provides a shared standard for web content accessibility that meets the needs of individuals, organizations, and governments internationally. [1] WCAG 2.2 is the latest iteration of these guidelines, building upon and extending the robust framework of its predecessors, WCAG 2.0 and WCAG 2.1.[1] The W3C explicitly advises the use of WCAG 2.2 to maximize the future applicability of accessibility efforts, positioning it as the current best practice. [2]

For Product Owners, understanding and implementing WCAG 2.2 is not merely a technical checkbox; it is a strategic imperative. These guidelines are designed to make web content accessible to a broad spectrum of individuals with disabilities. This includes, but is not limited to, those with visual impairments (such as blindness or low vision), auditory disabilities (deafness or hard of hearing), physical or motor limitations, speech disabilities, cognitive and learning challenges, and neurological conditions. [1] The evolution of WCAG to version 2.2 reflects a progressively deeper understanding of these diverse user needs. This version particularly enhances support for users with low vision, cognitive and learning disabilities, and motor disabilities, and notably provides increased guidance on mobile accessibility, acknowledging the diverse devices people use today. [5] This focus signifies that product development aligned with WCAG 2.2 is in step with contemporary accessibility thinking, potentially future-proofing products against evolving standards.

Adherence to WCAG 2.2 transcends basic compliance. It translates into tangible business benefits:

- **Expanded Market Reach:** By making products accessible, businesses can tap into a larger market segment, including the significant population of individuals

with disabilities and an aging population that often experiences age-related impairments.

- **Enhanced User Experience for All:** Many accessibility features improve usability for everyone, not just users with disabilities. For instance, clear contrast benefits users in bright sunlight, and captions help in noisy environments or when audio cannot be played.
- **Improved Brand Reputation:** A commitment to accessibility signals inclusivity and social responsibility, enhancing brand image and fostering customer loyalty. [5]
- **Legal and Regulatory Compliance:** Many jurisdictions worldwide have laws and policies mandating web accessibility, often referencing WCAG standards. Conformance with WCAG 2.2 Level AA typically allows organizations to meet or exceed these requirements, mitigating legal risks. [2] Products aiming only for older versions like WCAG 2.0 or 2.1 may be falling behind in addressing known accessibility gaps, potentially leading to a suboptimal user experience for certain disability groups and an increased risk of non-compliance as regulations evolve to reflect newer standards like WCAG 2.2. [2]

A crucial practical point for Product Owners is WCAG 2.2's backward compatibility. Web pages that conform to WCAG 2.2 also conform to WCAG 2.1 and WCAG 2.0 at the same level or higher. [3] This means that efforts invested in meeting WCAG 2.2 automatically cover the requirements of older standards, simplifying compliance narratives and ensuring that previous accessibility work remains relevant. This makes adopting WCAG 2.2 an upgrade, not a wholesale replacement, of accessibility efforts.

Section 1.2: Core Accessibility Concepts for Product Owners

To effectively champion accessibility, Product Owners should be familiar with the foundational concepts of WCAG. These concepts provide the framework for understanding and implementing the specific guidelines and success criteria.

The Four Principles of Accessibility (POUR)

At the heart of WCAG are four guiding principles, often remembered by the acronym POUR. These principles are the bedrock of web accessibility, ensuring that digital products are designed to be inclusive. [1]

1. **Perceivable:** Information and user interface components must be presentable to users in ways they can perceive. This means that content cannot be invisible to all of a user's senses. For example, images need text alternatives for those who cannot see them, and audio content needs captions for those who cannot hear. [1]

2. **Operable:** User interface components and navigation must be operable. The interface cannot require interaction that a user cannot perform. For instance, all functionality should be available from a keyboard for users who cannot use a mouse, and users must be given enough time to read and use content. [1]
3. **Understandable:** Information and the operation of the user interface must be understandable. The content or operation cannot be beyond a user's comprehension. This includes making text readable and understandable, and making web pages appear and operate in predictable ways.[1]
4. **Robust:** Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including current and future assistive technologies. As technologies and user agents evolve, the content should remain accessible. [1]

These POUR principles are not just abstract categories; they represent fundamental aspects of human-computer interaction. When these principles are addressed effectively, they often lead to a better user experience for *all* users, not solely those with disabilities. For example, a website with clear, logical navigation (Operable) and easy-to-understand language (Understandable) benefits everyone. Framing accessibility through POUR can help Product Owners articulate its value as a core component of good design and overall product quality, rather than a niche or compliance-only requirement.

Understanding Guidelines, Success Criteria, and Conformance Levels

WCAG is structured in layers of guidance to provide a clear path to accessibility [1]:

- **Guidelines:** Under each of the four principles, there are 13 guidelines. These guidelines provide the basic goals that authors should work toward to make content more accessible to users with different disabilities. They are not testable themselves but offer the framework and overall objectives to help understand the success criteria and implement techniques effectively. [1]
- **Success Criteria (SCs):** For each guideline, testable success criteria are provided. These are the specific, actionable, and verifiable requirements that must be met to conform to WCAG. This report will focus on detailing these success criteria. [1] The hierarchical structure from Principles to Guidelines to Success Criteria allows Product Owners to grasp the strategic "why" at the higher levels and then engage with their teams on the tactical "how" at the Success Criterion and technique level.
- **Conformance Levels (A, AA, AAA):** To meet the needs of different groups and situations, WCAG defines three levels of conformance for the success criteria [1]:

- **Level A (Lowest):** This level addresses the most basic and essential web accessibility features. Failing to meet these criteria can create significant barriers for users with disabilities, impacting a large number of users. [1]
- **Level AA (Mid-range):** This level deals with some of the biggest and most common barriers for disabled users. It is more comprehensive than Level A and is the target conformance level for most accessibility legislation and policies worldwide. Achieving Level AA ensures a good baseline of accessibility. [1]
- **Level AAA (Highest):** This level includes criteria for the highest level of web accessibility. It addresses more specific and nuanced issues, further enhancing the user experience for people with disabilities. However, it may not be possible or practical to satisfy all Level A/AA criteria for all types of content. [1]

Why Level A & AA are Key Targets for Product Development

This report focuses on WCAG 2.2 Level A and Level AA success criteria because they represent the most impactful and commonly targeted standards for product accessibility. Level AA conformance is generally the recommended and sought-after standard for most websites and applications. [2] Achieving this level ensures that a product addresses significant accessibility barriers and often aligns with legal and regulatory requirements. [4] The global consensus on Level AA as a reasonable and achievable standard provides Product Owners with a clear, defensible target for their accessibility efforts.

Part 2: The WCAG 2.2 Level A & AA Standards: Your Comprehensive Guide for Product Development

Introduction to the Table: How to Use This Guide

The following comprehensive table details each WCAG 2.2 Level A and Level AA success criterion. It is designed to provide Product Owners with a clear, actionable summary, translating technical standards into practical insights for product development.

Each row in the table represents a specific success criterion and is organized into the following columns:

1. **SC Number & Full Title (Conformance Level):** The official identifier, title, and conformance level (A or AA) of the success criterion.

2. **Description (In Layman's Terms):** A simple explanation of what the standard means and its core objective.
3. **Why It Matters (User Benefit & Product Impact):** Identifies who benefits from this standard (often highlighting specific disabilities but also broader user groups) and explains how meeting it can improve the product (e.g., wider audience, better SEO, enhanced usability, reduced legal risk).
4. **Key Considerations & Instructions for Product Owners:** Actionable advice and questions for Product Owners to consider and discuss with their design and development teams. This column aims to bridge the gap between understanding the standard and implementing it in the product lifecycle.
5. **Simple Examples:** One or two relatable scenarios illustrating the success criterion in action, often contrasting good and bad practices to clarify the concept.

This table is structured according to the four POUR principles (Perceivable, Operable, Understandable, Robust), then by their respective guidelines, and finally by each success criterion. This organization helps to understand how individual requirements contribute to broader accessibility goals.

The information presented is drawn from the official WCAG 2.2 documentation and its supporting "Understanding WCAG 2.2" documents. [1] These "Understanding" documents are the authoritative source for detailed interpretation and provide extensive guidance on how to meet each criterion. Product Owners should encourage their teams to consult these resources for in-depth information when specific questions or complex scenarios arise.

It's important to recognize that many success criteria are interconnected. For example, providing clear visual labels for form fields (SC 3.3.2) is fundamental. However, ensuring these labels are programmatically associated with the controls (SC 1.3.1) and that the accessible name used by assistive technologies matches the visible label (SC 2.5.3) makes them truly effective for users relying on such technologies. Accessibility is often an ecosystem of requirements working in concert.

The new success criteria introduced in WCAG 2.2 (e.g., 2.4.11 Focus Not Obscured (Minimum), 2.5.7 Dragging Movements, 2.5.8 Target Size (Minimum), 3.2.6 Consistent Help, 3.3.7 Redundant Entry, 3.3.8 Accessible Authentication (Minimum)) particularly address modern web interactions, mobile usability, and cognitive load. This reflects a clear trend towards making complex and dynamic digital products more accessible

and signals areas where Product Owners should focus attention, especially for new development.

Finally, several success criteria include specific "exceptions." While these exceptions provide flexibility in certain situations, Product Owners should be aware of them but also understand that relying on exceptions should be a carefully considered decision, not a default strategy. The overarching goal is to achieve maximum accessibility, and exceptions should only be invoked when truly essential and no other accessible solution is feasible.

Comprehensive Table of WCAG 2.2 Level A & AA Success Criteria

SC Number & Full Title (Conformance Level)	Description (In Layman's Terms)	Why It Matters (User Benefit & Product Impact)	Key Considerations & Instructions for Product Owners	Simple Examples
Principle 1: Perceivable – Information and user interface components must be presentable to users in ways they can perceive.				
Guideline 1.1 Text Alternatives: Provide text alternatives for any non-text content so that it can be changed into other forms				

people need, such as large print, braille, speech, symbols or simpler language.				
1.1.1 Nontext Content (Level A)	All images, icons, charts, and other non-text elements that convey information need a text description (alt text). Decorative elements should be marked so assistive tech can ignore them. [6]	**Users:** Blind users (screen readers announce alt text), low vision users (can enlarge text), people with cognitive disabilities (text can clarify complex images). **Product:** Improves SEO (search engines index text), content is usable if images fail to load. [6]	Ensure design/content teams provide meaningful alt text for all informative images. For controls (e.g., image buttons), alt text must describe the function. Discuss how purely decorative images will be handled by developers (e.g., null alt attribute).	**Good:** An image of a company logo has alt text "Acme Corp Logo." A photo of a team has alt text "The marketing team smiling in the office." **Bad:** An important chart showing sales data has alt text "chart.jpg" or no alt text.
Guideline 1.2 Time-based Media: Provide alternatives for time-based media.				
1.2.1 Audio-only and Video-only (Prerecorded) (Level A)	For audio-only content (like a podcast), a text transcript is needed. For	**Users:** Deaf/hard of hearing users (transcripts for audio), blind/low	If your product uses audio-only or video-only content, plan for creating	**Good:** A podcast episode page includes a link to a full text

	video-only content (like a silent animation), a text transcript or an audio track describing the visuals is needed. [1]	vision users (audio description or transcript for video-only). **Product:** Content becomes accessible to a wider audience, text versions are searchable. [7]	transcripts or audio descriptions. Ensure these alternatives are easy to find.	transcript. A silent animated tutorial has an option to play an audio track describing the actions. **Bad:** An audio lecture is provided with no transcript.
1.2.2 Captions (Prerecorded) (Level A)	All prerecorded videos with audio (e.g., product demos, training videos) must have synchronized captions. [1]	**Users:** Deaf/hard of hearing users (can read dialogue and important sounds), people in noisy environments, non-native speakers. **Product:** Increases video engagement, improves comprehension, content is usable in more situations. [8]	Budget and plan for caption creation for all prerecorded video content. Ensure captions are accurate and synchronized.	**Good:** A marketing video has accurate, timed captions for all speech and important sound effects (e.g., "[door slams]"). **Bad:** A training video has auto-generated captions that are full of errors and not synchronized.
1.2.3 Audio Description or Media Alternative (Prerecorded) (Level A)	For videos with important visual information not conveyed in the main audio (e.g., actions, scene changes), either provide audio description	**Users:** Blind/low vision users (understand visual context), people with cognitive disabilities (may better understand with	Review videos: does the existing audio fully explain what's happening visually? If not, plan for audio description or a	**Good:** A movie scene shows a character silently looking for a clue. Audio description says, "Sarah scans the room, her eyes landing

	(narrating these visuals) or a full text transcript that includes descriptions of visuals. [1]	added narration or text). **Product:** Makes video content fully accessible, provides richer understanding for all. [9]	comprehensive media alternative (like a detailed script).	on a small, wooden box under the bed." Or, a text transcript describes this scene. **Bad:** A product demo shows features being used without any spoken explanation of the visual steps.
1.2.4 Captions (Live) (Level AA)	All live audio content in synchronized media (e.g., live webinars, streaming events) must have captions. [10]	**Users:** Deaf/hard of hearing users can participate in live events. **Product:** Makes live events inclusive, expands audience reach for real-time content. [11]	If your product involves live broadcasts or webinars, arrange for live captioning services (e.g., CART). This often requires advance planning and budget.	**Good:** A live-streamed company announcement has real-time captions displayed. **Bad:** A live Q&A webinar has no captions, excluding deaf/hard of hearing participants.
1.2.5 Audio Description (Prerecorded) (Level AA)	For all prerecorded videos, audio description must be provided for important visual content not covered by the main audio track. This is a stricter version	**Users:** Blind/low vision users (get a full understanding of visual information), some cognitive disabilities (helps interpret visual info). **Product:** Ensures full	Prioritize audio description for key video content. This involves scripting and recording additional narration that fits into natural pauses in the existing audio.	**Good:** A training video includes an optional audio track where a narrator describes on-screen actions, charts, and text that isn't spoken by the main presenter. **Bad:**

	of 1.2.3, requiring audio description specifically, not just a media alternative. [1]	accessibility of video content for visually impaired users. [12]		A complex instructional video relies heavily on visuals with no audio description provided.
Guideline 1.3 Adaptable: Create content that can be presented in different ways (for example simpler layout) without losing information or structure.				
1.3.1 Info and Relationships (Level A)	The way information is structured and related visually (e.g., headings, lists, tables, form field groupings, bold text for emphasis) must also be understandable by assistive technologies. This is usually done through correct HTML markup. [1]	**Users:** Screen reader users (understand page structure, navigate by headings/tables), users who customize page appearance (structure is maintained). **Product:** Improves content organization, enhances usability for assistive tech, can improve SEO. [13]	Ensure developers use semantic HTML (e.g., <h1>-<h6> for headings, <ul>/<ol>/<li> for lists, <table> with <th> for tables). Don't rely on visual styling alone to convey structure (e.g., just making text bigger for a heading).	**Good:** A form groups related fields (e.g., "Billing Address") using a <fieldset> and <legend>. Headings are marked with <h1>, <h2>, etc. **Bad:** A list of items is created using line breaks and asterisks instead of <ul> and <li> tags. Required fields are only indicated by red color.

1.3.2 Meaningful Sequence (Level A)	If the order in which content is presented affects its meaning, that correct reading order must be programmatically determinable (e.g., by a screen reader). [1]	**Users:** Screen reader users (hear content in a logical order), keyboard-only users (tab through content logically). **Product:** Ensures content makes sense regardless of presentation changes (e.g., CSS styling, assistive tech rendering). [14]	Discuss with developers how content order is managed, especially when CSS is used for visual layout (e.g., multi-column layouts). The underlying code order should generally reflect the logical reading order.	**Good:** A news article with a main story and a sidebar. Even if the sidebar appears first visually due to CSS, the HTML source order places the main story first, so screen readers read it correctly. **Bad:** A form's fields are visually arranged logically, but the HTML source order is jumbled, causing screen readers to announce fields out of sequence.
1.3.3 Sensory Characteristics (Level A)	Instructions for understanding or operating content should not rely *only* on sensory details like shape, color, size, visual location, orientation, or sound. [1]	**Users:** Blind/low vision users (cannot rely on visual cues), colorblind users (cannot distinguish based on color alone). **Product:** Makes instructions universally understandable. [10]	Review all instructions. If they say "Click the round blue button on the right," ensure there's another way to identify that button (e.g., "Click the 'Submit' button, which is round, blue, and on the right").	**Good:** "To proceed, press the 'Next' button (shaped like an arrow, located in the bottom right corner)." **Bad:** "Press the square button to continue." (If there are multiple square buttons). "Items in red require immediate

				attention." (Without any other indicator).
1.3.4 Orientation (Level AA)	Content should not be locked to a specific display orientation (portrait or landscape) unless a specific orientation is essential for its use. [15]	**Users:** Users with devices mounted in a fixed orientation (e.g., on a wheelchair), low vision users (may prefer landscape for larger text). **Product:** Ensures usability on various devices and for users with specific setup needs, crucial for mobile accessibility. [16]	Ensure your web pages and applications adapt to both portrait and landscape views. Avoid programmatically restricting orientation.	**Good:** A news website displays correctly in both portrait and landscape mode on a tablet. **Bad:** A mobile app forces users into landscape mode even for reading text articles. **Exception:** A banking app requiring landscape mode to accurately scan a check.
1.3.5 Identify Input Purpose (Level AA)	The purpose of form fields collecting information *about the user* (e.g., name, address, email) should be programmatically identifiable, often using the HTML autocomplete attribute. [1]	**Users:** People with cognitive/memory disabilities (benefit from autofill), motor impairments (reduces typing), users of assistive tech (can present familiar icons or customized prompts). **Product:** Speeds up form completion,	For forms collecting common user data, work with developers to implement the autocomplete attribute with appropriate values (e.g., autocomplete="given-name", autocomplete="email").	**Good:** A checkout form uses autocomplete attributes for name, address, and credit card fields, allowing the browser to offer autofill suggestions. **Bad:** A registration form asks for name and email but doesn't use autocomplete,

		reduces errors, improves user experience, especially for repeat users. [17]		forcing users to type everything manually each time.
Guideline 1.4 Distinguishable: Make it easier for users to see and hear content including separating foreground from background.				
1.4.1 Use of Color (Level A)	Color should not be the *only* visual means of conveying information, indicating an action, prompting a response, or distinguishing an element. [1]	**Users:** Colorblind users (may not perceive color differences), low vision users (may not distinguish colors well), users of monochrome screens. **Product:** Ensures critical information is accessible to a wider range of visual abilities. [18]	If color is used to convey meaning (e.g., red for errors, green for success), ensure there's an additional visual cue (e.g., an icon, bold text, an asterisk) or text explanation.	**Good:** Required form fields are marked with red text *and* an asterisk (*). A graph uses different colors *and* different patterns for its lines. **Bad:** A list of links uses only color to differentiate visited from unvisited links.
1.4.2 Audio Control (Level A)	If any audio on a webpage plays automatically	**Users:** Screen reader users (auto-playing	Avoid auto-playing audio. If it's essential,	**Good:** A website has background

	for more than 3 seconds, there must be a way for the user to pause or stop it, or control its volume independently of the system volume. [1]	audio can interfere with screen reader output), users with attention difficulties (can be highly distracting). **Product:** Prevents user frustration and ensures users can consume content without unwanted audio interference. [19]	ensure clear, easily accessible controls (pause/stop/volume) are provided *before* the audio starts or very early on.	music that starts automatically but provides a visible "Pause Music" button at the top of the page. **Bad:** A promotional video with sound auto-plays as soon as the page loads, with no obvious way to stop it.
1.4.3 Contrast (Minimum) (Level AA)	Text and images of text must have a contrast ratio of at least 4.5:1 against their background. Large text (18pt or 14pt bold) needs at least 3:1. Exceptions for logos and decorative text. [1]	**Users:** People with low vision, colorblindness, older users with reduced contrast sensitivity. **Product:** Improves readability for a wide range of users in various lighting conditions. [15]	Work with designers to choose color palettes that meet these contrast ratios. Use contrast checking tools during design and development. This applies to all text, including placeholder text in forms.	**Good:** Dark grey text on a light grey background with a 5:1 contrast ratio. Large headlines in medium grey on a white background with a 3.5:1 ratio. **Bad:** Light grey text on a white background (e.g., 2:1 contrast).
1.4.4 Resize Text (Level AA)	Text must be resizable up to 200% without loss of content or functionality, and without requiring	**Users:** People with low vision (can enlarge text to make it readable). **Product:** Ensures content	Ensure developers use relative units (like ems or percentages) for text sizes and container	**Good:** A user zooms their browser to 200%, and all text increases in size, and lines wrap correctly

	assistive technology. [20]	remains usable when users adjust browser zoom or text size settings. [21]	widths, rather than fixed pixel values, to allow text to reflow properly when resized. Test by zooming the browser to 200%.	within their containers. No text is cut off or overlaps. **Bad:** When text is resized, it overflows its container, becomes unreadable, or causes horizontal scrollbars to appear for lines of text.
1.4.5 Images of Text (Level AA)	Use actual text instead of images of text whenever possible. If an image of text must be used, it's only allowed if the presentation is essential (e.g., a logo) or if the image of text can be customized by the user. [1]	**Users:** People with low vision (cannot easily resize or change colors of text in images), screen reader users (text in images may not be read or read well if alt text is poor). **Product:** Text is more flexible, searchable, and adaptable. Reduces reliance on potentially problematic images. [22]	Instruct design and content teams to use real text styled with CSS for headings, navigation, and other textual elements. Reserve images of text for true exceptions like logos.	**Good:** Website headings are styled using CSS to achieve a specific font and appearance. **Bad:** Navigation menu items are created as individual images of text. A paragraph of important information is presented as a JPEG.
1.4.10 Reflow (Level AA)	Content must be viewable without needing to scroll in two dimensions	**Users:** People with low vision who use screen magnification (avoid	This heavily ties into responsive design. Ensure layouts adapt to narrow	**Good:** When a webpage is viewed on a mobile phone or zoomed to

	(horizontally and vertically) when zoomed to 400% (equivalent to a 320 CSS pixel width viewport for vertical scrolling content, or 256 CSS pixel height for horizontal scrolling content). Exceptions for content requiring 2D layout for meaning (e.g., maps, complex data tables, video). [1]	disorienting bi-directional scrolling). **Product:** Improves usability on small screens (mobile responsiveness) and for magnified views. [23]	viewports, typically by reflowing content into a single column. Test by resizing the browser window or using developer tools to simulate small viewports.	400%, the text wraps, and images scale down, so the user only needs to scroll vertically. **Bad:** At 400% zoom, the user must scroll horizontally to read lines of text and vertically to move down the page.
1.4.11 Nontext Contrast (Level AA)	Visual information used to identify user interface components (like buttons, input borders) and important parts of graphics (like lines in a chart) must have a contrast ratio of at least 3:1 against adjacent colors. Exceptions for inactive components or when	**Users:** People with low vision (can better distinguish UI controls and graphical information). **Product:** Improves the clarity and usability of interactive elements and informative graphics. [24]	Designers must ensure that UI elements (e.g., input borders, checkbox outlines, icons acting as controls) and essential parts of graphics have sufficient contrast. This is distinct from text contrast (1.4.3).	**Good:** An input field has a visible border with a 3:1 contrast against the background. A pie chart uses colors for its slices that have 3:1 contrast with each other, or has borders between slices that meet this contrast. **Bad:** A button's border is very light and blends into the background. Icons in a

	appearance is browser-determined and unmodified. [1]			diagram are too faint to distinguish.
1.4.12 Text Spacing (Level AA)	Users must be able to override author-set text spacing (line height, paragraph spacing, letter spacing, word spacing) to certain minimum values without loss of content or functionality. [1]	**Users:** People with low vision or dyslexia (can adjust spacing to improve readability). **Product:** Content remains usable even if users apply custom stylesheets or browser extensions to modify text spacing. [25]	Developers should build content so it doesn't break if users apply the specified spacing overrides (e.g., text shouldn't get cut off or overlap). Avoid using fixed heights on containers of text.	**Good:** A user applies a bookmarklet to increase line height and word spacing. The text reflows correctly, and no content is lost or becomes unreadable. **Bad:** When a user increases line height, text gets cut off at the bottom of its container.
1.4.13 Content on Hover or Focus (Level AA)	If additional content appears on mouse hover or keyboard focus (e.g., tooltips, submenus), it must be: 1. **Dismissible** (e.g., via Esc key, without moving hover/focus), 2. **Hoverable** (pointer can move over the new content), and 3. **Persistent**	**Users:** Low vision users (need time and ability to perceive magnified pop-up content), users with motor impairments (may accidentally trigger pop-ups or need to move pointer onto them). **Product:** Makes interactive pop-ups less	For custom tooltips, dropdown menus, or other pop-ups triggered by hover/focus: ensure they can be closed with Esc, that the mouse can move onto them without them disappearing, and that they don't vanish too quickly.	**Good:** A custom tooltip appears on hover. The user can move their mouse over the tooltip to read it, and it stays visible. Pressing Esc closes it. **Bad:** A mega menu appears on hover but disappears if the user tries to move their mouse down into the menu options. A

	(stays visible until trigger removed, dismissed, or info invalid). Exceptions for user-agent controlled content (like native browser tooltips) or non-obscuring input errors. [1]	frustrating and more usable, especially for magnified views or keyboard navigation. [20]		tooltip is so small and disappears so fast that a magnified view makes it impossible to read.
Principle 2: Operable – User interface components and navigation must be operable.				
Guideline 2.1 Keyboard Accessible: Make all functionality available from a keyboard.				
2.1.1 Keyboard (Level A)	All functionality of the content must be operable through a keyboard interface without requiring specific timings for individual	**Users:** Blind users (rely on keyboard), users with motor impairments (cannot use a mouse), some low vision users (find keyboard easier).	Ensure all interactive elements (links, buttons, form fields, custom widgets) can be reached and activated using only the keyboard (e.g., Tab, Shift+Tab,	**Good:** A user can navigate through a form, select options in a dropdown, and submit the form using only the Tab, Arrow, and Enter keys. A drag-and-drop interface

	keystrokes. Exception for functionality that inherently requires path-dependent input (e.g., freehand drawing). [26]	**Product:** Essential for a large group of users with disabilities; enables use of various assistive technologies that emulate keyboard input. [27]	Enter, Spacebar, Arrow keys). Test thoroughly.	also offers "cut" and "paste" keyboard commands. **Bad:** A custom date picker can only be operated with a mouse.
2.1.2 No Keyboard Trap (Level A)	If keyboard focus can be moved to a component, it must be possible to move focus *away* from that component using only the keyboard. If non-standard keys are needed (beyond Tab/Arrows/Esc), the user must be advised. [1]	**Users:** Keyboard-only users (prevents them from getting stuck in a part of the page). **Product:** Avoids major usability dead-ends and frustration for keyboard navigators. [28]	Test all interactive components, especially third-party widgets or plugins. Ensure that after tabbing into a component, the user can tab out or use Esc to return to the main page flow.	**Good:** A modal dialog traps focus within itself (which is okay for modals), but provides "OK" and "Cancel" buttons that can be activated by keyboard to close it, returning focus to the page. **Bad:** A user tabs into an embedded media player, and the Tab key then only cycles within the player's controls, with no way to tab out to the rest of the page.
2.1.4 Character Key Shortcuts (Level A)	If single character keys (letters,	**Users:** Speech input users (speaking words	Avoid single-character shortcuts if	**Good:** A webmail application uses

	numbers, punctuation, symbols) are used as shortcuts, at least one of these must be true: 1. The shortcut can be turned off. 2. The shortcut can be remapped to include a modifier key (Ctrl, Alt, etc.). 3. The shortcut is only active when the component it relates to has focus. [1]	can accidentally trigger single-key shortcuts), users with motor impairments (prone to accidental key presses). **Product:** Prevents accidental activation of commands, making the interface more predictable and less error-prone for these users. [29]	possible. If used, provide settings to disable or remap them, or ensure they only work when a specific input field or widget has focus.	'j' and 'k' to navigate messages but only when the message list has focus. It also offers a setting to disable these shortcuts. **Bad:** Pressing 'm' anywhere on a page mutes all audio, and this cannot be turned off or changed.
Guideline 2.2 Enough Time: Provide users enough time to read and use content.				
2.2.1 Timing Adjustable (Level A)	For any time limit set by the content, at least one is true: user can turn it off, adjust it (to at least 10x default), extend it (with 20+ sec warning, at least 10 extensions), or it's a real-	**Users:** People with cognitive disabilities, learning disabilities, motor impairments, visual impairments (all may need more time to read, comprehend, or	Avoid unnecessary time limits. If session timeouts are needed for security, provide clear warnings and easy ways to extend the session. For content that	**Good:** An online banking session warns the user 2 minutes before timeout and provides a button to extend the session by 15 minutes. A news ticker allows users to pause

	time/essential exception, or it's >20 hours. [1]	interact). **Product:** Prevents users from being "timed out" of critical tasks, reduces frustration, makes services accessible to those who need more time. [30]	updates (e.g., carousels), allow users to pause or adjust timing.	the scrolling. **Bad:** An online quiz has a strict 30-second limit per question with no way to pause or extend.
2.2.2 Pause, Stop, Hide (Level A)	For any moving, blinking, or scrolling information that starts automatically, lasts >5 seconds, and is parallel with other content, there must be a user control to pause, stop, or hide it (unless essential). For auto-updating info, provide controls to pause, stop, hide, or adjust frequency (unless essential). [1]	**Users:** People with attention deficit disorders (can be highly distracting), cognitive/learning disabilities (difficulty reading with moving content), screen reader users (moving content can interfere). **Product:** Improves focus, reduces distraction, makes page usable for those sensitive to motion. [26]	Avoid auto-playing animations, carousels, or blinking content. If used, ensure they stop after 5 seconds or provide clear pause/stop controls. This is critical for non-interference.	**Good:** An animated banner ad stops moving after 5 seconds. A content carousel has clear "pause," "next," and "previous" buttons. **Bad:** A website has a constantly scrolling news ticker at the top of every page with no way to stop it.
Guideline 2.3 Seizures and Physical Reactions: Do not design				

content in a way that is known to cause seizures or physical reactions.				
2.3.1 Three Flashes or Below Threshold (Level A)	Web pages must not contain anything that flashes more than three times in any one-second period, or the flash must be below general flash and red flash thresholds. [31]	**Users:** People with photosensitive epilepsy (prevents triggering seizures). **Product:** Critical for user safety; avoids causing serious physical harm. [32]	This is a strict requirement. Ensure no content (videos, animations, GIFs) violates flash thresholds. Use tools to test flashing content if unsure.	**Good:** A video game scene with explosions ensures that flashing effects are small enough or infrequent enough to pass the thresholds. **Bad:** A website uses a rapidly flashing banner advertisement.
Guideline 2.4 Navigable: Provide ways to help users navigate, find content, and determine where they are.				
2.4.1 Bypass Blocks (Level A)	A mechanism must be available to bypass blocks of content that are repeated on multiple web pages (e.g., navigation menus,	**Users:** Keyboard-only users and screen reader users (can skip repetitive navigation and get to main content faster). **Product:**	Implement a "skip to main content" link as the first focusable item on pages with recurring navigation blocks. Using heading	**Good:** A website has a "Skip to main content" link at the top of each page that becomes visible on keyboard focus. **Bad:** Users must tab

	headers). [1]	Improves navigation efficiency, reduces frustration for sequential navigators. [33]	structures properly also helps.	through 50 navigation links on every page to reach the primary content.
2.4.2 Page Titled (Level A)	Web pages must have titles that describe their topic or purpose. [1]	**Users:** All users (helps identify content in browser tabs, bookmarks, search results), especially screen reader users (title is often the first thing announced). **Product:** Improves findability, orientation, and SEO.[34]	Ensure every page has a unique and descriptive <title> element in the HTML. For Single Page Applications (SPAs), update the title dynamically as the view changes.	**Good:** A product page title is "Product Name - Category - Company Name." A contact page is titled "Contact Us - Company Name." **Bad:** All pages on a site have the title "Homepage" or are just the URL.
2.4.3 Focus Order (Level A)	If a page can be navigated sequentially (e.g., by keyboard tabbing) and the navigation order affects meaning or operation, focusable components must receive focus in an order that preserves meaning and	**Users:** Keyboard-only users (expect logical tab order), screen reader users (logical order aids comprehension) . **Product:** Ensures predictable and understandable navigation flow, especially for complex forms	Ensure the default tab order (based on HTML source) is logical. If tabindex is used to modify focus order, do so carefully to maintain a sensible path. Test tabbing through all interactive elements.	**Good:** In a form, tabbing moves sequentially from First Name, to Last Name, to Email, etc. **Bad:** Tabbing through a form jumps from First Name to Zip Code, then back to Email, then to City.

	operability. [1]	or interfaces. [35]		
2.4.4 Link Purpose (In Context) (Level A)	The purpose of each link must be clear from its link text alone, or from the link text together with its programmatically determined context (e.g., surrounding sentence, list item, table cell/header). Exception for links where purpose is ambiguous to all users (e.g., in a game). [1]	**Users:** Screen reader users (can understand link destination from a list of links), users with cognitive/motor impairments (avoid clicking wrong links). **Product:** Improves navigation clarity, reduces user errors, enhances usability. [31]	Write descriptive link text. Avoid generic phrases like "Click here" or "Read more" unless the surrounding context clearly explains the destination.	**Good:** "Read more about our Q3 financial results." Link text: "Q3 financial results." **Bad:** Multiple links on a page all say "Read more" with no other context provided directly with the link.
2.4.5 Multiple Ways (Level AA)	More than one way must be available to locate a web page within a set of web pages (e.g., site search, site map, navigation menu, table of contents), unless the page is the result or step in a process (e.g., checkout confirmation). [36]	**Users:** Accommodates different user preferences and abilities for finding content (e.g., some prefer search, others prefer browsing a menu). **Product:** Improves content discoverability and site usability for a wider range of	For most websites, ensure users have options like a primary navigation menu AND a site search. A site map can also fulfill this.	**Good:** A university website has a main navigation menu, a search bar, and an A-Z index of departments. **Bad:** A large corporate website only has a complex hierarchical navigation menu with no search function.

		users. [37]		
2.4.6 Headings and Labels (Level AA)	Headings (if present) and labels (if present) must describe the topic or purpose of the content or control they are associated with. [1]	**Users:** All users (understand content structure and form field purpose), especially screen reader users (navigate by headings, understand form controls via labels). **Product:** Improves content organization, scannability, and form usability. [38]	Ensure headings clearly indicate the content of their section. Ensure form field labels clearly describe the expected input. This is about the descriptiveness of the text, not just its presence or markup.	**Good:** A section about payment options is titled "Payment Options." A form field for entering an email address is labeled "Email Address." **Bad:** A section heading is vague like "Information." A form field is labeled "Input1."
2.4.7 Focus Visible (Level AA)	Any keyboard-operable user interface must have a mode of operation where the keyboard focus indicator is visible. [1]	**Users:** Sighted keyboard-only users (need to see where their focus is), users with attention/cognitive limitations (helps track focus). **Product:** Essential for keyboard navigation; makes it clear which element will receive keyboard input.	Ensure that when users tab through interactive elements, a clear visual indicator (e.g., an outline, border, background change) shows which element currently has focus. Do not remove or suppress the browser's default focus indicator unless	**Good:** When a button receives keyboard focus, a distinct blue outline appears around it. **Bad:** CSS is used to set outline: none; on all focusable elements, making the focus invisible.

		[39]	a more visible custom one is provided.	
2.4.11 Focus Not Obscured (Minimum) (Level AA) **(New in 2.2)**	When a UI component receives keyboard focus, it must not be *entirely* hidden by author-created content (e.g., sticky headers/footers, non-modal dialogs, cookie banners). [1]	**Users:** Sighted keyboard users (need to see the focused element to interact), low vision users (obscured elements are harder to find/use). **Product:** Prevents frustrating situations where focused elements are hidden, improving keyboard usability, especially with modern sticky/overlay patterns. [41]	Test keyboard navigation with any sticky elements or pop-ups. Ensure the focused item is at least partially visible. Consider using scroll padding or ensuring non-modal dialogs don't completely cover focused items.	**Good:** A page has a sticky footer. When tabbing to a link that would be under the footer, the page scrolls slightly so the link is visible above the footer. **Bad:** A cookie banner at the bottom of the screen completely covers a link when it receives focus, and the page doesn't scroll to reveal it.
Guideline 2.5 Input Modalities: Make it easier for users to operate functionality through various inputs beyond keyboard.				

2.5.1 Pointer Gestures (Level A)	All functionality that uses multipoint (e.g., pinch-to-zoom) or path-based (e.g., swipe, drag in a specific path) gestures must also be operable with a single pointer without a path-based gesture, unless the gesture is essential. [1]	**Users:** Users with motor impairments (may not be able to perform complex gestures), users of alternative input devices (head pointers, eye gaze may not support these gestures). **Product:** Ensures features are usable by people who cannot perform complex touch/pointer gestures. [36]	If using pinch-zoom, swiping carousels, or path-dependent sliders, provide alternative single-click/tap controls (e.g., +/- buttons for zoom, arrow buttons for carousels, direct input fields for sliders).	**Good:** A map allows pinch-to-zoom but also has "+" and "-" buttons for zooming. A slider can be dragged but also allows clicking on the track to set a value. **Bad:** A photo gallery can only be navigated by swiping left/right.
2.5.2 Pointer Cancellation (Level A)	For functionality operated by a single pointer, at least one is true: 1. No down-event triggers the action. 2. Action is completed on up-event, with a way to abort before completion (e.g., drag off target). 3. Up-event reverses any down-event outcome. 4. Down-event	**Users:** Users prone to accidental clicks/taps (motor impairments, cognitive disabilities, or anyone). **Product:** Reduces accidental actions, makes interface more forgiving and predictable. [43]	Design interactions so actions trigger on "mouse up" or "touch end" inside the target. Allow users to drag their finger/pointer away from a target before releasing to cancel an action.	**Good:** A button activates when the user releases the mouse button *while the pointer is still over the button* . If they press down, drag off, then release, nothing happens. **Bad:** A button activates the instant the user presses down on it, with no way to cancel if it was

	activation is essential. [42]			an accidental press.
2.5.3 Label in Name (Level A)	For UI components with visible text labels (or images of text as labels), the component's programmatic "accessible name" (used by assistive tech) must contain the visible label text. [1]	**Users:** Speech input users (can activate controls by saying the visible label), screen reader users (what they hear matches what sighted users see). **Product:** Creates consistent and predictable interaction for voice control and screen reader users. [44]	Ensure the text content of a <label> element is part of the accessible name of the input it's for. If using aria-label or aria-labelledby, the accessible name must include the visible label text. Best practice: start the accessible name with the visible label text.	**Good:** A button is visibly labeled "Search". Its accessible name is also "Search". Another button is "Submit Order"; its accessible name is "Submit Order Now". **Bad:** A button is labeled "Go" but its accessible name (e.g., via aria-label) is "Start process". Speech users saying "Go" won't activate it.
2.5.4 Motion Actuation (Level A)	Functionality operated by device motion (e.g., shaking, tilting) or user motion (e.g., waving at a camera) must also be operable by UI components (e.g., buttons), and the motion activation can be disabled (unless motion is essential or via an	**Users:** Users with motor impairments (cannot perform motions), users with mounted devices. Prevents accidental activation for users with tremors. **Product:** Makes motion-based features accessible to more users and prevents	If you have "shake to undo" or "tilt to scroll," provide on-screen buttons for "Undo" or scrolling. Offer a setting to disable motion detection.	**Good:** A music app allows skipping tracks by shaking the device but also has on-screen "Next" and "Previous" buttons. There's a setting to turn off "shake to skip." **Bad:** A game requires tilting the device to steer, with no alternative controls.

	accessibility-supported interface). [1]	accidental triggers. [45]		
2.5.7 Dragging Movements (Level AA) **(New in 2.2)**	All functionality that uses a dragging movement (e.g., drag-and-drop, sliders) must also be achievable by a single pointer without dragging, unless dragging is essential or user-agent controlled. [1]	**Users:** People with motor impairments or using alternative input devices (trackballs, head pointers, speech-controlled mice) who find dragging difficult or impossible. **Product:** Makes drag-based interactions accessible to a wider range of users who can use a pointer but not drag. [46]	For drag-and-drop list reordering, provide up/down arrow buttons for selected items. For sliders, allow clicking on the track or provide input fields to set the value.	**Good:** A file upload interface allows dragging files but also has a "Browse" button. A slider can be dragged, but also has buttons to increment/decrement its value. **Bad:** A Kanban board only allows moving tasks between columns by dragging and dropping.
2.5.8 Target Size (Minimum) (Level AA) **(New in 2.2)**	The size of targets for pointer inputs (e.g., buttons, links) must be at least 24 by 24 CSS pixels. Exceptions: if undersized targets are adequately spaced, if an equivalent larger control exists, if inline with text, if	**Users:** People with motor impairments (tremors, limited dexterity), users on touchscreens (especially mobile), users in unstable environments (e.g., on a train). **Product:** Reduces accidental clicks/taps,	Aim for interactive targets to be at least 24x24 CSS pixels. If smaller, ensure sufficient spacing around them so a 24px diameter circle centered on each target doesn't overlap others.	**Good:** Icon buttons in a toolbar are all 30x30 CSS pixels. Small "x" close icons are 18x18px but have enough surrounding space. **Bad:** A row of tiny 16x16px icon buttons are packed tightly together with no

	user-agent controlled and unmodified, or if size is essential/legally required. [1]	improves usability and reduces frustration, especially on touch devices. [42]		spacing.
Principle 3: Understandable – Information and the operation of user interface must be understandable.				
Guideline 3.1 Readable: Make text content readable and understandable.				
3.1.1 Language of Page (Level A)	The default human language of each web page must be programmatically identifiable (e.g., using the lang attribute on the <html> element). [47]	**Users:** Screen reader users (enables correct pronunciation), browser users (can display characters correctly, offer translation). **Product:** Ensures correct rendering and pronunciation by assistive tech and browsers,	Set the lang attribute on the <html> tag for every page (e.g., <html lang="en">).	**Good:** An English website has <html lang="en">. A French website has <html lang="fr">. **Bad:** A webpage has no lang attribute, or it's incorrect (e.g., lang="us" instead of lang="en-US" or lang="en").

		aids internationalization. [48]		
3.1.2 Language of Parts (Level AA)	The human language of any passage or phrase within the content that is in a different language from the page's default must be programmatically identifiable. Exceptions for proper names, technical terms, vernacular words. [1]	**Users:** Screen reader users (correct pronunciation of foreign words/phrases), users of braille displays (correct character representation). **Product:** Improves comprehension for multilingual content. [49]	If a paragraph is in English but contains a French phrase, wrap the French phrase in a with a lang="fr" attribute.	**Good:** On an English page: "His motto was Carpe diem." **Bad:** An English page quotes a sentence in Spanish with no language attribute, causing screen readers to mispronounce it.
Guideline 3.2 Predictable: Make Web pages appear and operate in predictable ways.				
3.2.1 On Focus (Level A)	When any UI component receives focus, it must not initiate a change of context (e.g., automatically submitting a form, launching a new window, or moving focus to another	**Users:** Keyboard users, screen reader users, users with cognitive/attention difficulties (prevents unexpected and disorienting changes). **Product:**	Ensure that merely tabbing to or clicking into a field does not trigger an action. Actions should occur on explicit activation (e.g., pressing Enter, clicking a button).	**Good:** A user tabs to a dropdown menu. The menu might open, but the page doesn't navigate away until the user selects an option and presses Enter. **Bad:** Tabbing

	component). [1]	Creates a more stable and predictable user experience. [50]		into a radio button automatically submits the form.
3.2.2 On Input (Level A)	Changing the setting of any UI component (e.g., checking a box, typing in a field, selecting from a list) must not automatically cause a change of context unless the user has been advised of the behavior *before* using the component. [1]	**Users:** Users with visual disabilities, cognitive limitations (prevents unexpected page changes or submissions they aren't prepared for). **Product:** Enhances predictability and user control over actions. [51]	Avoid controls that automatically submit forms or navigate to new pages when their value changes. If such behavior is necessary, clearly inform the user beforehand (e.g., "Selecting a state will refresh the page to show local stores").	**Good:** A user selects an option from a dropdown list, and nothing happens until they click a "Submit" button. Or, a dropdown says "Select your country (page will reload)". **Bad:** Selecting an option from a dropdown list immediately navigates to a new URL without warning.
3.2.3 Consistent Navigation (Level AA)	Navigational mechanisms (e.g., menus, links) that are repeated on multiple web pages within a set must occur in the same relative order each time they are repeated, unless a change is initiated by the user. [1]	**Users:** All users, especially those with cognitive limitations, low vision, or who are blind (helps predict where to find navigation elements). **Product:** Improves site learnability and navigation efficiency. [47]	Maintain a consistent order for main navigation links, footer links, breadcrumbs, etc., across all pages of your site or application section.	**Good:** The main navigation menu (Home, About, Services, Contact) appears in that same order at the top of every page. **Bad:** The "Contact Us" link is sometimes in the header, sometimes in the footer, and

				sometimes in a sidebar on different pages.
3.2.4 Consistent Identification (Level AA)	Components that have the same functionality within a set of web pages must be identified consistently (e.g., same label, icon, or alt text). [52]	**Users:** Screen reader users (recognize functions by consistent names), users with cognitive limitations (reduces confusion). **Product:** Improves predictability and learnability of the interface. [53]	If an icon (e.g., a printer icon) or a button label (e.g., "Submit") is used for the same function across multiple pages, ensure its accessible name (and visual label if applicable) is consistent.	**Good:** An icon of a shopping cart is used on all pages to link to the shopping cart and always has the alt text "Shopping Cart." **Bad:** A button to submit a form is labeled "Send" on one page and "Submit Query" on another, though it does the same thing.
3.2.6 Consistent Help (Level A) **(New in 2.2)**	If help mechanisms (e.g., contact info, FAQs, chat) are repeated on multiple pages within a set, they must occur in the same relative order to other page content, unless a change is initiated by the user. [1]	**Users:** Users who need assistance, especially those with cognitive disabilities (makes help easier to find consistently). **Product:** Improves user support, reduces frustration, helps users complete tasks. [54]	If you have a "Help" link, "Contact Us" page, or embedded chat widget, ensure it's located in a consistent place in the page structure (e.g., always in the header, always in the footer) across pages.	**Good:** A link to the "FAQ" page is always the last item in the footer menu on every page of the website. **Bad:** The "Live Chat" option is in the header on the homepage, in a sidebar on product pages, and in the footer on the contact page.
Guideline 3.3				

Input Assistance: Help users avoid and correct mistakes.				
3.3.1 Error Identification (Level A)	If an input error is automatically detected, the item in error must be identified, and the error described to the user in text. [1]	**Users:** All users, especially those who are blind or color-deficient (can perceive errors via text), users with cognitive/learning disabilities (helps understand what's wrong). **Product:** Enables users to understand and fix errors, reducing form abandonment. [55]	When a form submission fails, clearly indicate which fields have errors and provide a text description of each error (e.g., "Email address is missing," "Password must be at least 8 characters").	**Good:** After submitting a form with a missing required field, the field is highlighted, and a message "Username is required" appears next to it. **Bad:** A form simply reloads with no indication of what went wrong, or only highlights erroneous fields in red without any text explanation.
3.3.2 Labels or Instructions (Level A)	Labels or instructions must be provided when content requires user input. [1]	**Users:** All users, especially those with cognitive/learning disabilities or screen reader users (understand what input is expected). **Product:** Improves form usability and	Ensure all form fields (text inputs, checkboxes, radio buttons, select menus) have clear, visible labels. Provide instructions for any specific formatting requirements	**Good:** A text field for a phone number is labeled "Phone Number". A password field has instructions: "Must be at least 8 characters and include a number." **Bad:** A form has several

		accuracy of data entry. [56]	(e.g., "Date (MM/DD/YYYY)") .	text fields with no visible labels, relying only on placeholder text (which disappears on input).
3.3.3 Error Suggestion (Level AA)	If an input error is automatically detected and suggestions for correction are known, these suggestions must be provided to the user (unless it jeopardizes security or purpose). [1]	**Users:** Users with learning disabilities, visual impairments, motor impairments (helps them correct errors more easily). **Product:** Reduces user frustration, improves task completion rates, makes forms more helpful. [52]	If a user misspells a city name, suggest corrections. If they enter a date in the wrong format, explain the correct format. If a username is taken, suggest alternatives.	**Good:** User types "Decmber" for month. Error message: "Invalid month. Did you mean December? Or choose from: January, February..." **Bad:** User enters an invalid email format. Error message: "Invalid email." (No suggestion on what a valid format looks like).
3.3.4 Error Prevention (Legal, Financial, Data) (Level AA)	For web pages causing legal commitments, financial transactions, modifying/deleting user-controllable data, or submitting test responses, at least one is true: actions are	**Users:** All users, especially those prone to errors (e.g., due to dyslexia, motor impairments) to avoid serious, irreversible consequences. **Product:** Prevents costly mistakes, builds user trust,	For online purchases, allow users to review their order before final submission. For deleting important data, provide a confirmation step ("Are you sure?"). Allow reversal of	**Good:** Before finalizing an online purchase, a summary page shows all items, costs, and shipping info, with an "Edit Order" button. **Bad:** Clicking "Delete Account" immediately and

	reversible, input is checked for errors and user can correct, or a mechanism exists to review, confirm, and correct before finalizing. [1]	reduces support issues. [57]	actions where feasible.	irreversibly deletes all user data without a confirmation step.
3.3.7 Redundant Entry (Level A) **(New in 2.2)**	Information previously entered by or provided to the user that is required again in the same process must be either auto-populated or available for the user to select. Exceptions for essential re-entry, security, or invalid previous info. [1]	**Users:** People with cognitive/memory difficulties (reduces recall burden), motor impairments (reduces typing). **Product:** Speeds up multi-step processes, reduces user effort and errors, improves completion rates. [58]	In a checkout process, if shipping and billing addresses are requested, provide a checkbox "Billing address same as shipping" to auto-fill. If a username is entered on step 1, pre-fill it if asked again on step 3 of the same process.	**Good:** During a multi-page job application, if the user enters their name on page 1, and page 3 asks for it again, the field is pre-filled with the name from page 1. **Bad:** A user fills out their delivery address, then on the next page for billing, has to type it all out again even if it's identical.
3.3.8 Accessible Authentication (Minimum) (Level AA) **(New in 2.2)**	A cognitive function test (e.g., remembering a password, solving a puzzle) is not required for any step in an authentication process unless an alternative	**Users:** People with memory impairments, dyslexia, dyscalculia, or other cognitive disabilities who find password recall or puzzle solving difficult/impossible. **Product:**	Support password managers (don't block autofill). Allow copy-pasting into password fields. Offer alternatives like "Log in with Google/Apple" or WebAuthn	**Good:** A website allows users to log in using a username/password (and supports password manager autofill) OR by using their Google account.

	method not relying on a cognitive test is offered, or a mechanism assists (e.g., password manager support, copy-paste), or it's object/personal content recognition. [1]	Makes login accessible to more users, supports use of password managers, can improve security by allowing more complex (but managed) passwords. [59]	(biometrics, security keys) if feasible. Avoid CAPTCHAs that are purely cognitive tests without accessible alternatives.	Two-factor authentication allows using a security key instead of typing a code. **Bad:** A website requires users to memorize and type a complex password and solve a visual puzzle CAPTCHA, with no alternatives or assistance mechanisms.
Principle 4: Robust – Content must be robust enough that it can be interpreted by a wide variety of user agents, including assistive technologies.				
Guideline 4.1 Compatible: Maximize compatibility with current and future user agents, including assistive technologies.				

4.1.2 Name, Role, Value (Level A)	For all UI components (forms, links, scripted components), their name and role must be programmatically determinable; states, properties, and values settable by the user must be programmatically settable; and changes to these must be available to assistive tech. [1]	**Users:** Screen reader users and users of other assistive tech (AT can understand and interact with controls). **Product:** Essential for AT compatibility; enables users with disabilities to operate custom widgets and complex interfaces. [60]	If using standard HTML controls correctly, this is often met. For custom components (e.g., built with s and JavaScript), use WAI-ARIA attributes to define name (e.g., aria-label), role (e.g., role="button"), and states/properties (e.g., aria-expanded, aria-checked).	**Good:** A custom-built dropdown menu uses ARIA roles (button, listbox, option) and properties (aria-haspopup, aria-expanded) so screen readers announce it correctly. **Bad:** A clickable element is styled to look like a button but has no role="button" or keyboard operability, making it inaccessible to screen readers and keyboard users.
4.1.3 Status Messages (Level AA)	Status messages (e.g., "Search results updated," "Item added to cart," "Form error") must be programmatically determinable through role or properties so assistive tech can present them to the user *without* the message needing to	**Users:** Screen reader users (are informed of important changes or feedback that appear on screen without shifting focus). **Product:** Provides timely feedback to users of assistive tech, improving their awareness of application	Use ARIA live regions (e.g., role="status", aria-live="polite") or specific alert roles (role="alert") for content that updates dynamically to provide feedback.	**Good:** When a user adds an item to the cart, a message "Item added to cart" appears on the page and is announced by screen readers, even though focus remains on the "Add to Cart" button. **Bad:** A "Loading..." message appears visually

	receive focus. [1]	state. [61]		but isn't announced by screen readers, leaving them unsure if anything is happening.

Part 3: What's New and Different in WCAG 2.2: A Product Owner's Briefing

WCAG 2.2 is not a complete overhaul but an important evolution, refining existing concepts and introducing new success criteria to address contemporary web design challenges and a deeper understanding of user needs. For Product Owners, understanding these changes is key to ensuring products remain at the forefront of accessibility.

Section 3.1: Overview of New Level A and AA Success Criteria in WCAG 2.2

WCAG 2.2 introduces several new success criteria at Level A and AA, signifying their broad applicability and importance for a baseline of modern accessibility. [1] These new criteria often target areas like mobile usability, interactions for users with cognitive or motor impairments, and common patterns in modern web applications. The introduction of these SCs at Level A and AA by the W3C signals that these are not niche concerns but fundamental to current accessibility best practices. Product Owners should therefore prioritize understanding and implementing these in both new and existing products.

Here is a concise summary of the new Level A and AA success criteria:

- **2.4.11 Focus Not Obscured (Minimum) (Level AA):** This criterion ensures that when a user interface component receives keyboard focus, it is not entirely hidden by other content created by the author, such as sticky headers or footers, or non-modal dialogs. [1] It addresses a common frustration for keyboard users where the focused element is not visible, making interaction difficult or impossible.
    - *Problem Solved:* Prevents focused interactive elements from being completely hidden by persistent on-screen elements like cookie banners or fixed navigation bars, which is crucial for sighted keyboard navigators.
- **2.5.7 Dragging Movements (Level AA):** This requires that any functionality

using a dragging movement (like drag-and-drop or sliders) can also be operated by a single pointer without dragging, unless dragging is essential.[1]
    - *Problem Solved:* Assists users who can use a pointer but find dragging actions difficult or impossible due to motor impairments or the nature of their input device (e.g., head pointers, eye-gaze systems).
- **2.5.8 Target Size (Minimum) (Level AA):** This standard mandates a minimum target size of 24 by 24 CSS pixels for pointer inputs, or sufficient spacing around smaller targets, to make them easier to activate accurately. [1]
    - *Problem Solved:* Reduces accidental activation of adjacent controls, benefiting users with motor impairments and anyone using a touchscreen, especially on mobile devices where precision can be challenging.
- **3.2.6 Consistent Help (Level A):** If help mechanisms (like contact details, FAQs, or chat support) are provided and repeated on multiple pages, they must appear in the same relative order each time. [1]
    - *Problem Solved:* Makes it easier for users, particularly those with cognitive disabilities or those in distress, to find help consistently across a website or application.
- **3.3.7 Redundant Entry (Level A):** This criterion aims to reduce user effort by requiring that information previously entered by the user within the same process is either auto-populated or made available for selection if needed again. [1]
    - *Problem Solved:* Lessens cognitive load and typing effort, especially beneficial for users with memory or motor impairments, by avoiding the need to repeatedly enter the same information in a multi-step process.
- **3.3.8 Accessible Authentication (Minimum) (Level AA):** This success criterion stipulates that authentication processes must not require a cognitive function test (like remembering a password or solving a puzzle) unless an accessible alternative is provided or a mechanism is available to assist the user. [1] (Note: Some sources incorrectly list this as Level A, but the official W3C documentation and more reliable summaries confirm it as Level AA [1]).
    - *Problem Solved:* Supports users with cognitive disabilities (e.g., memory impairments, dyslexia) by ensuring they can log in without being solely reliant on memory-intensive tasks. It encourages support for password managers and alternative authentication methods.

The focus of these new criteria on issues like focus visibility with sticky elements, touch target sizes, alternatives to dragging, consistent help, minimizing redundant data entry, and more accessible authentication methods demonstrates WCAG's responsiveness to how digital products are built and used today. They directly

address common pain points, particularly for mobile users and individuals with cognitive, motor, or low vision disabilities.[5]

Section 3.2: Understanding Removed Criteria: The Case of 4.1.1 Parsing

WCAG 2.2 has removed one success criterion that was present in previous versions: SC 4.1.1 Parsing[4]. For Product Owners, it's useful to understand why such changes occur.

SC 4.1.1 Parsing originally required that content implemented using markup languages have complete start and end tags, elements nested according to their specifications, no duplicate attributes, and unique IDs, to ensure that assistive technologies could parse the content reliably.

The removal of 4.1.1 Parsing reflects the evolution of web technologies and browser capabilities.[3] Modern web browsers have become significantly more adept at handling minor parsing errors in HTML. The issues that 4.1.1 aimed to prevent – where malformed code could break assistive technology interpretation – are now less common or are often covered by other success criteria, such as 4.1.2 Name, Role, Value, which necessitates robust code for components to be programmatically determinable.[60] Furthermore, SC 4.1.1 was found to be challenging to test consistently, and its failures did not always correlate with actual accessibility barriers for users, especially if other good practices were followed.

The removal of this criterion simplifies compliance testing to some extent. However, Product Owners should reassure their teams that this does not diminish the importance of writing well-formed, valid HTML. Adhering to web standards and producing clean code remains a best practice and is foundational to the "Robust" principle of accessibility – ensuring content can be reliably interpreted by a wide variety of user agents, including assistive technologies. The removal of 4.1.1 is an example of WCAG adapting, removing criteria that become redundant or less impactful due to technological advancements, rather than simply accumulating rules.

Part 4: Actionable Next Steps: Embedding Accessibility into Your Product Strategy

Understanding the WCAG 2.2 Level A and AA standards is the first step. The next, crucial phase for Product Owners is to actively embed these principles and requirements into the product development lifecycle and team culture.

Section 4.1: Integrating WCAG 2.2 into the Product Development Lifecycle

Addressing accessibility late in the development cycle is often costly, time-consuming, and can lead to compromised solutions. A far more effective approach is to "shift left," incorporating accessibility considerations from the very beginning of the product lifecycle – during ideation, design, and development, rather than treating it as a final compliance check. Product Owners are pivotal in championing this proactive integration.

Practical ways Product Owners can drive this include:

- **Including Accessibility in Definitions:** Incorporate accessibility requirements directly into user stories and acceptance criteria. For example, "As a keyboard-only user, I want to be able to complete the checkout process using only my keyboard, so that I can purchase products independently." Acceptance criteria could then specify relevant SCs like 2.1.1 Keyboard and 2.4.7 Focus Visible.
- **Early Design Considerations:** Ensure that designers are aware of and apply WCAG 2.2 principles from the outset. This includes considerations for color contrast (1.4.3, 1.4.11), focus indicators (2.4.7, 2.4.11), target sizes (2.5.8), logical layout for reflow (1.4.10), and clear navigation patterns.
- **Development Best Practices:** Work with development teams to ensure they are using semantic HTML (1.3.1), implementing ARIA correctly for custom components (4.1.2), and testing for keyboard accessibility (2.1.1) and other operable criteria throughout development.
- **Dedicated Testing Time:** Allocate specific time within sprints for accessibility testing, including both automated tools (which can catch some issues) and manual testing (essential for many criteria, especially those related to operability and understandability).
- **QA Integration:** Collaborate with Quality Assurance teams to include accessibility checks in their standard test plans. This ensures that accessibility is treated as a core quality attribute of the product.
- **Advocacy for Training:** Champion and secure resources for accessibility training for all team members (designers, developers, content creators, QA). A knowledgeable team is better equipped to build accessible products natively.

The benefits of this proactive approach are significant. It leads to reduced rework, as accessibility issues are caught and addressed early when they are easier and cheaper to fix. It fosters a more inclusive product from the ground up, leading to higher user satisfaction, broader market reach, and enhanced brand reputation. [5] By making

accessibility a core part of the product definition and development process, Product Owners can significantly influence the creation of truly usable and inclusive digital experiences.

Section 4.2: Key Questions for Product Owners to Ask Their Teams

Product Owners do not need to be accessibility engineers, but they can significantly drive accessibility by asking pertinent and insightful questions throughout the product lifecycle. These questions can stimulate critical thinking within the team and ensure that accessibility is a continuous consideration.

Here are some key questions, aligned with WCAG principles and common challenges, that Product Owners can ask their design, development, and QA teams:

Regarding Perceivability:

- "How will users who are blind access the information conveyed in this image/chart/video?" (Relates to SC 1.1.1, 1.2.1, 1.2.2, 1.2.3, 1.2.5)
- "Have we checked all text and important non-text elements (like icons and input borders) for sufficient color contrast against their backgrounds, considering users with low vision or color blindness?" (Relates to SC 1.4.3, 1.4.11)
- "Can users easily resize the text on this page to 200% without content breaking or becoming unreadable?" (Relates to SC 1.4.4)
- "Does our layout reflow properly on small screens or when zoomed, so users don't have to scroll in two directions?" (Relates to SC 1.4.10)

Regarding Operability:

- "How will a user who cannot use a mouse operate this feature? Is all functionality keyboard accessible?" (Relates to SC 2.1.1)
- "Have we tested for keyboard traps, especially in complex widgets or modals?" (Relates to SC 2.1.2)
- "Is our keyboard focus indicator always visible and clear?" (Relates to SC 2.4.7)
- "Are any focused elements being hidden by sticky headers, footers, or pop-ups?" (Relates to SC 2.4.11)
- "If we use swipe gestures or drag-and-drop, what are the single-pointer alternatives for users who can't perform those actions?" (Relates to SC 2.5.1, 2.5.7)
- "Are our buttons and interactive targets large enough or adequately spaced for easy activation, especially on touchscreens?" (Relates to SC 2.5.8)

Regarding Understandability:

- "Is the language of the page and any foreign language phrases correctly identified for assistive technologies?" (Relates to SC 3.1.1, 3.1.2)
- "Are our navigation menus and interactive components consistent across the site/application?" (Relates to SC 3.2.3, 3.2.4)
- "Is help (like FAQs or contact information) easy to find and consistently located?" (Relates to SC 3.2.6)
- "Are all form fields clearly labeled? Are instructions provided for complex inputs?" (Relates to SC 3.3.2)
- "When a user makes an error in a form, is the error clearly identified in text, and do we provide helpful suggestions for correction?" (Relates to SC 3.3.1, 3.3.3)
- "For critical actions like purchases or data deletion, how are we helping users prevent errors or confirm their intent?" (Relates to SC 3.3.4)
- "Are we asking users to re-enter information they've already provided in this process? Can we auto-fill it or make it selectable?" (Relates to SC 3.3.7)
- "How are we ensuring our login process is accessible, especially for users who might struggle with remembering passwords or solving puzzles?" (Relates to SC 3.3.8)

Regarding Robustness:

- "For any custom-built components, how are we ensuring they correctly communicate their name, role, and state to assistive technologies like screen readers?" (Relates to SC 4.1.2)
- "When the page updates dynamically (e.g., search results load, a status changes), how are screen reader users informed of these changes if focus doesn't move?" (Relates to SC 4.1.3)

By asking these types of questions, Product Owners can foster a culture of accessibility awareness and accountability within their teams. This proactive engagement helps ensure that accessibility is not an afterthought but an integral part of creating high-quality, inclusive digital products that serve all users effectively.

Conclusions and Recommendations

The Web Content Accessibility Guidelines (WCAG) 2.2 provide an essential framework for creating digital products that are accessible to and usable by the widest possible audience, including individuals with diverse disabilities. For Product Owners, embracing WCAG 2.2 Level A and AA standards is not merely a compliance exercise;

it is a strategic approach to enhance user experience, expand market reach, mitigate legal risks, and build a more inclusive brand.

Key Conclusions:

1. **WCAG 2.2 is the Current Standard:** The W3C recommends WCAG 2.2 for all new and updated web content, as it addresses a broader range of accessibility needs, particularly for users with cognitive, learning, and motor disabilities, and offers improved guidance for mobile accessibility. [2] Its backward compatibility ensures that meeting 2.2 also satisfies requirements for 2.0 and 2.1. [3]
2. **Accessibility is Holistic:** The POUR principles (Perceivable, Operable, Understandable, Robust) underscore that accessibility is about fundamental usability for all, not just a niche requirement. [1] Many success criteria are interconnected, requiring a comprehensive approach rather than isolated fixes.
3. **New Criteria Address Modern Challenges:** The new success criteria in WCAG 2.2 (Focus Not Obscured, Dragging Movements, Target Size, Consistent Help, Redundant Entry, Accessible Authentication) directly tackle common issues in modern web design, such as sticky elements, touch interactions, and cognitive load in multi-step processes. [5]
4. **Product Owner Role is Pivotal:** Product Owners are uniquely positioned to champion accessibility by integrating it into the product vision, strategy, and development lifecycle from the earliest stages.

Actionable Recommendations for Product Owners:

1. **Prioritize WCAG 2.2 Level AA Conformance:** Make WCAG 2.2 Level AA the target standard for your digital products. This level provides a strong baseline of accessibility and aligns with most international legal and policy requirements.
2. **Educate and Empower Your Team:** Advocate for and invest in accessibility training for designers, developers, content creators, and QA testers. Familiarize them with the POUR principles and the specific Level A and AA success criteria relevant to their roles.
3. **Integrate Accessibility into the Workflow ("Shift Left"):**
   - Include accessibility requirements in user stories and acceptance criteria.
   - Ensure design mockups and prototypes consider accessibility (e.g., color contrast, focus states, target sizes).
   - Allocate time for both automated and manual accessibility testing throughout the development process.
   - Make accessibility a standard part of QA testing.

4. **Leverage the "Understanding WCAG 2.2" Documents:** While this guide provides a Product Owner-focused summary, encourage teams to consult the official "Understanding WCAG 2.2" documents for detailed technical guidance and techniques for each success criterion. [1]
5. **Focus on New WCAG 2.2 Criteria:** Pay particular attention to the newly introduced Level A and AA success criteria, as they address modern web interactions and previously underserved user needs. Ensure these are incorporated into new development and plans for updating existing products.
6. **Ask Probing Questions:** Regularly engage your teams with questions about how accessibility is being addressed for different user needs and scenarios, using the examples provided in this guide as a starting point.
7. **Test with Real Users:** Whenever possible, include users with disabilities in your user testing processes. Their firsthand feedback is invaluable for identifying real-world accessibility barriers that automated tools or checklists might miss.
8. **View Accessibility as an Ongoing Commitment:** Accessibility is not a one-time project but an ongoing process of improvement and maintenance as new content is added, features are updated, and standards evolve.

By adopting these recommendations, Product Owners can lead their teams in creating digital products that are not only compliant but also genuinely inclusive, usable, and valuable to all users. This commitment ultimately contributes to better products, a stronger brand, and a more equitable digital world.

Works cited

1. Web Content Accessibility Guidelines (WCAG) 2.2 - W3C, accessed on June 4, 2025, https://www.w3.org/TR/WCAG22/
2. Web Accessibility Guidelines - WCAG 2 Checklist - Clym, accessed on June 4, 2025, https://clym.io/accessibility-news/wcag-2-checklist
3. Requirements for Web Content Accessibility Guidelines 2.2 - W3C on GitHub, accessed on June 4, 2025, https://w3c.github.io/wcag/requirements/22/
4. WCAG Compliance | Guide | AudioEye®, accessed on June 4, 2025, https://www.audioeye.com/compliance/wcag/
5. WCAG 2.2: New Success Criteria, More Inclusive Content, accessed on June 4, 2025, https://www.wcag.com/blog/wcag-2-2-aa-summary-and-checklist-for-website-owners/
6. Understanding Success Criterion 1.1.1: Non-text Content | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/non-text-content.html
7. Understanding Success Criterion 1.2.1: Audio-only and Video-only ..., accessed on

June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/audio-only-and-video-only-prerecorded.html
8. Understanding Success Criterion 1.2.2: Captions (Prerecorded ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/captions-prerecorded.html
9. Understanding Success Criterion 1.2.3: Audio Description or Media ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/audio-description-or-media-alternative-prerecorded.html
10. Understanding Success Criterion 1.3.3: Sensory Characteristics ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/sensory-characteristics.html
11. Understanding Success Criterion 1.2.4: Captions (Live) | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/captions-live.html
12. Understanding Success Criterion 1.2.5: Audio Description ... - W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/audio-description-prerecorded.html
13. Understanding Success Criterion 1.3.1: Info and Relationships | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/info-and-relationships.html
14. Understanding Success Criterion 1.3.2: Meaningful Sequence | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/meaningful-sequence.html
15. Understanding Success Criterion 1.4.3: Contrast (Minimum) | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/contrast-minimum.html
16. Understanding Success Criterion 1.3.4: Orientation | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/orientation.html
17. Understanding Success Criterion 1.3.5: Identify Input Purpose | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/identify-input-purpose.html
18. Understanding Success Criterion 1.4.1: Use of Color | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/use-of-color.html
19. Understanding Success Criterion 1.4.2: Audio Control | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/audio-control.html
20. Understanding Success Criterion 1.4.13: Content on Hover or Focus ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/content-on-hover-or-focus.html
21. Understanding Success Criterion 1.4.4: Resize Text | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/resize-text.html
22. Understanding Success Criterion 1.4.5: Images of Text | WAI | W3C, accessed on

June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/images-of-text.html
23. Understanding Success Criterion 1.4.10: Reflow | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/reflow.html
24. Understanding Success Criterion 1.4.11: Non-text Contrast | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html
25. Understanding Success Criterion 1.4.12: Text Spacing | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/text-spacing.html
26. Understanding Success Criterion 2.2.2: Pause, Stop, Hide | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/pause-stop-hide.html
27. Understanding Success Criterion 2.1.1: Keyboard | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/keyboard.html
28. Understanding Success Criterion 2.1.2: No Keyboard Trap | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/no-keyboard-trap.html
29. Understanding Success Criterion 2.1.4: Character Key Shortcuts ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/character-key-shortcuts.html
30. Understanding Success Criterion 2.2.1: Timing Adjustable | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/timing-adjustable.html
31. Understanding Success Criterion 2.4.4: Link Purpose (In Context ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/link-purpose-in-context.html
32. Understanding Success Criterion 2.3.1: Three Flashes or Below ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/three-flashes-or-below-threshold.html
33. Understanding Success Criterion 2.4.1: Bypass Blocks | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/bypass-blocks.html
34. Understanding Success Criterion 2.4.2: Page Titled | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/page-titled.html
35. Understanding Success Criterion 2.4.3: Focus Order | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/focus-order.html
36. Understanding Success Criterion 2.5.1: Pointer Gestures | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/pointer-gestures.html
37. Understanding Success Criterion 2.4.5: Multiple Ways | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/multiple-

ways.html
38. Understanding Success Criterion 2.4.6: Headings and Labels | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/headings-and-labels.html
39. Understanding Success Criterion 2.4.7: Focus Visible | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/focus-visible.html
40. What's New in 2.2? - Digital Accessibility Office - Carnegie Mellon University, accessed on June 4, 2025, https://www.cmu.edu/computing/dao/news/2023/nov/wcag-2.2.html
41. Understanding Success Criterion 2.4.11: Focus Not Obscured ... - W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/focus-not-obscured-minimum.html
42. Understanding Success Criterion 2.5.8: Target Size (Minimum) | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum.html
43. Understanding Success Criterion 2.5.2: Pointer Cancellation | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/pointer-cancellation.html
44. Understanding Success Criterion 2.5.3: Label in Name | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/label-in-name.html
45. Understanding Success Criterion 2.5.4: Motion Actuation | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/motion-actuation.html
46. Understanding Success Criterion 2.5.7: Dragging Movements | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/dragging-movements.html
47. Understanding Success Criterion 3.2.3: Consistent Navigation | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/consistent-navigation.html
48. Understanding Success Criterion 3.1.1: Language of Page | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/language-of-page.html
49. Understanding Success Criterion 3.1.2: Language of Parts | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/language-of-parts.html
50. Understanding Success Criterion 3.2.1: On Focus | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/on-focus.html
51. Understanding Success Criterion 3.2.2: On Input | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/on-input.html
52. Understanding Success Criterion 3.3.3: Error Suggestion | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/error-

suggestion.html
53. Understanding Success Criterion 3.2.4: Consistent Identification ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/consistent-identification.html
54. Understanding Success Criterion 3.2.6: Consistent Help | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/consistent-help.html
55. Understanding Success Criterion 3.3.1: Error Identification | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/error-identification.html
56. Understanding Success Criterion 3.3.2: Labels or Instructions | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/labels-or-instructions.html
57. Understanding Success Criterion 3.3.4: Error Prevention (Legal ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/error-prevention-legal-financial-data.html
58. Understanding Success Criterion 3.3.7: Redundant Entry | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/redundant-entry.html
59. Understanding Success Criterion 3.3.8: Accessible Authentication ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/accessible-authentication-minimum.html
60. Understanding Success Criterion 4.1.2: Name, Role, Value | WAI ..., accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/name-role-value.html
61. Understanding Success Criterion 4.1.3: Status Messages | WAI | W3C, accessed on June 4, 2025, https://www.w3.org/WAI/WCAG22/Understanding/status-messages.html